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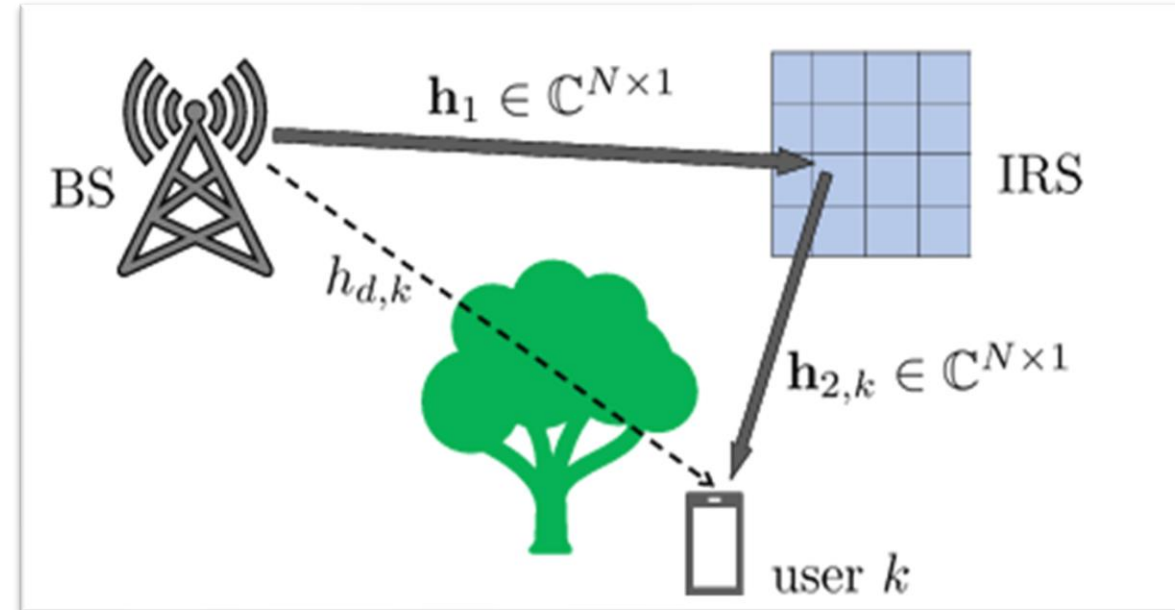
On the Impact of IRS on Out-of-Band Performance in Wireless Communications

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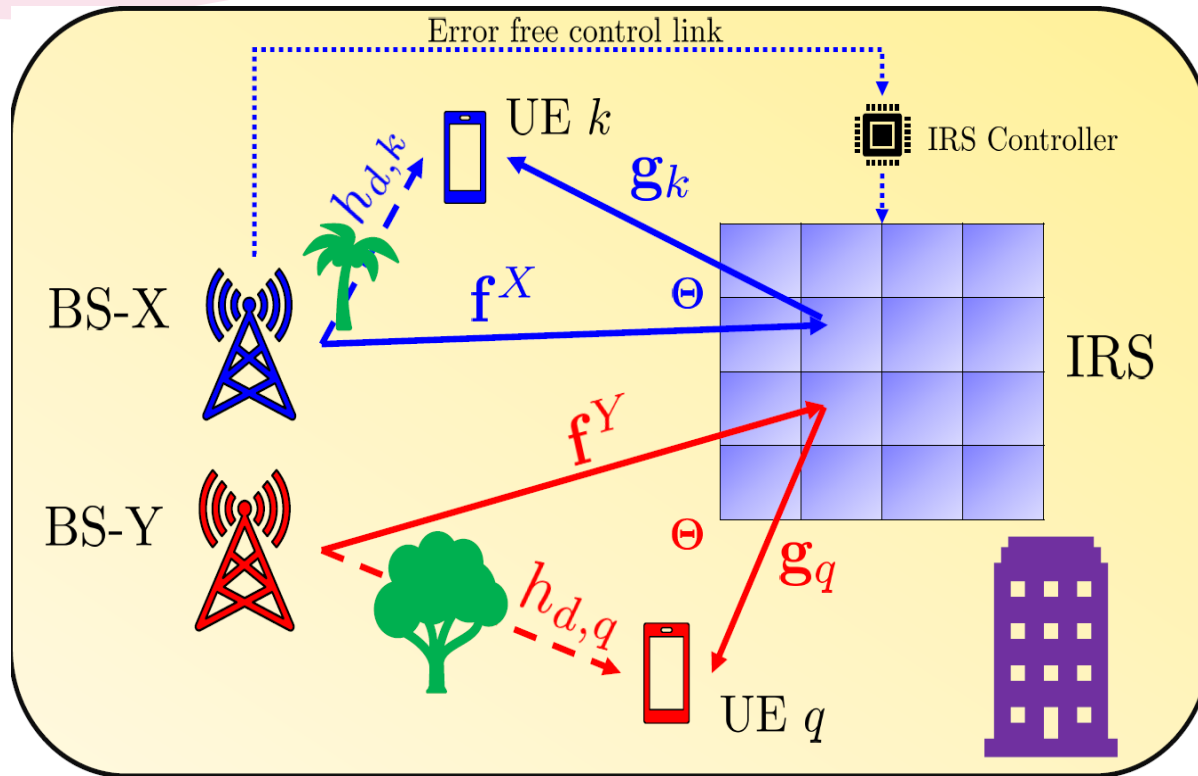
AUGUST 11, 2023

Introduction to IRS

- **Intelligent Reflecting Surfaces (IRS):** Meta surfaces with passive elements
 - **Alters** the overall channel - "*reflect the signals in specific directions.*"
- Perks
 - Boosts SINR
 - Energy efficiency
 - Coverage enhancement, etc.
- State-of-the art "**requirements**"
 - **CSI estimation** - complexity of $O(N)$
 - **Expensive** phase optimization
 - **Transportation** of phase angles: BS to IRS



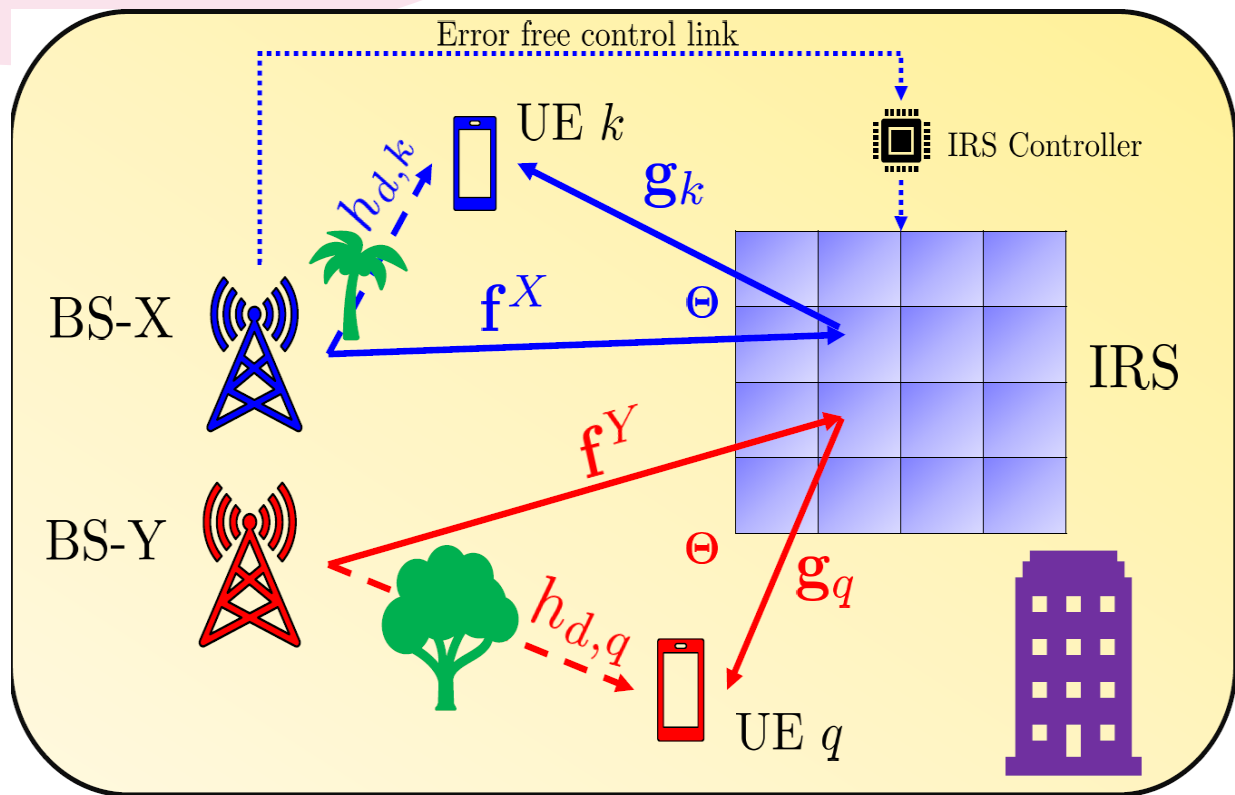
Motivation for the problem



- **NO** "practical" IRS can simultaneously beamform to UEs of BS - **X** & **Y**
 - Different channels @ in-band & OOB UEs
 - IRS is frequency dependent!
- Operator **X** controls the IRS
 - Optimal IRS @ in-band UEs
 - Random IRS @ OOB UEs

Does an IRS Degrade Out-of-Band Performance ?

System Setting



- Downlink signal at k -th UE (by BS-X):

$$y_k = (h_{d,k} + \mathbf{g}_k^T \mathbf{\Theta} \mathbf{f}^X) x_k + n_k,$$

- Downlink signal at q -th UE (by BS-Y):

$$y_q = (h_{d,q} + \mathbf{g}_q^T \mathbf{\Theta} \mathbf{f}^Y) x_q + n_q,$$

➡ Two operators: X and Y

➡ BS - X serves K UEs

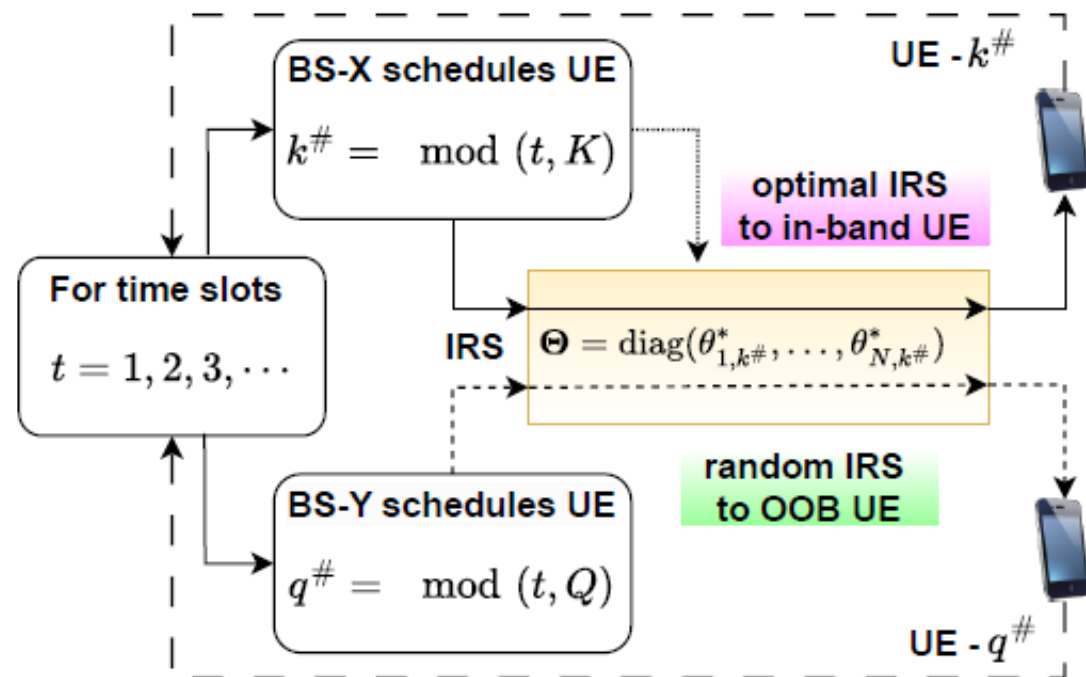
➡ BS - Y serves Q UEs

➡ Operator X controls the IRS

Scheduling Protocol: Round-Robin

Algorithm Protocol at BS-X and BS-Y

- 1: **Input:** UE indices at BS-X: $0, 1, \dots, K - 1$.
- 2: **Input:** UE indices at BS-Y: $0, 1, \dots, Q - 1$.
- 3: Initialize the UE index at BS-X to 0, at BS-Y to 0.
- 4: **repeat**
- 5: **for** time slot t **do**
- 6: $k^\# = \text{mod}(t, K)$. ▷ RR Scheduling at BS-X.
- 7: $q^\# = \text{mod}(t, Q)$. ▷ RR Scheduling at BS-Y.
- 8: Set $\Theta = \text{diag}(\theta_{1,k^\#}^*, \dots, \theta_{N,k^\#}^*)$ optimally.
- 9: ▷ Hence, a random Θ gets realized for OOB UEs.
- 10: Data transmission from BS-X to $k^\#$.
- 11: Data transmission from BS-Y to $q^\#$.
- 12: **end for**
- 13: **until** service time



OOB Performance in Sub-6 GHz Bands*

- ❑ **FR-1** Bands in the 5G standards
(410 MHz - 6 GHz, Rel. 15, 2018)
- ❑ **Rich-scattering** environment
- ❑ Channels follow independent **Rayleigh** fading



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The Benchmark

The SE of UE- k served by BS - X and UE - q served by BS - Y in an N -element IRS-aided system under the *simultaneous* beamforming is

$$R_k^{BF} = \log_2 \left(1 + \frac{P}{\sigma^2} \left| \sum_{n=1}^N |f_n^X g_{k,n}| + |h_{d,k}| \right|^2 \right),$$
$$R_q^{BF} = \log_2 \left(1 + \frac{P}{\sigma^2} \left| \sum_{n=1}^N |f_n^Y g_{q,n}| + |h_{d,q}| \right|^2 \right),$$

with the *beamforming* configuration of the *fictitious* IRS being

$$\theta_{n,k}^* = \angle h_{d,k} - (\angle f_n^X + \angle g_{k,n}), \quad n = 1, \dots, N,$$
$$\theta_{n,q}^* = \angle h_{d,q} - (\angle f_n^Y + \angle g_{q,n}), \quad n = 1, \dots, N.$$

- Ergodic SE at UE - k :

$$\langle R_k^{(X)} \rangle = \mathcal{E} \left[\log_2 \left(1 + \left| \sum_{n=1}^N |f_n^X g_{k,n}| + |h_{d,k}| \right|^2 \frac{P}{\sigma^2} \right) \right]$$

- Ergodic SE at UE - q :

$$\begin{aligned} \langle R_q^{(Y)} \rangle &= \mathcal{E} \left[\log_2 \left(1 + \left| \sum_{n=1}^N f_n^Y g_{q,n} e^{j\theta_n} + h_{d,q} \right|^2 \frac{P}{\sigma^2} \right) \right] \\ &= \mathcal{E} \left[\log_2 \left(1 + \left| \sum_{n=1}^N f_n^Y g_{q,n} + h_{d,q} \right|^2 \frac{P}{\sigma^2} \right) \right] \end{aligned}$$

- Sum Ergodic SE of operator - X & Y :

$$\bar{R}^{(X)} \triangleq \frac{1}{K} \sum_{k=1}^K \langle R_k^{(X)} \rangle \quad \& \quad \bar{R}^{(Y)} \triangleq \frac{1}{Q} \sum_{q=1}^Q \langle R_q^{(Y)} \rangle$$

Main Result - 1 (On Ergodic performance)

The ergodic sum SEs of operator **X** & **Y** scale as

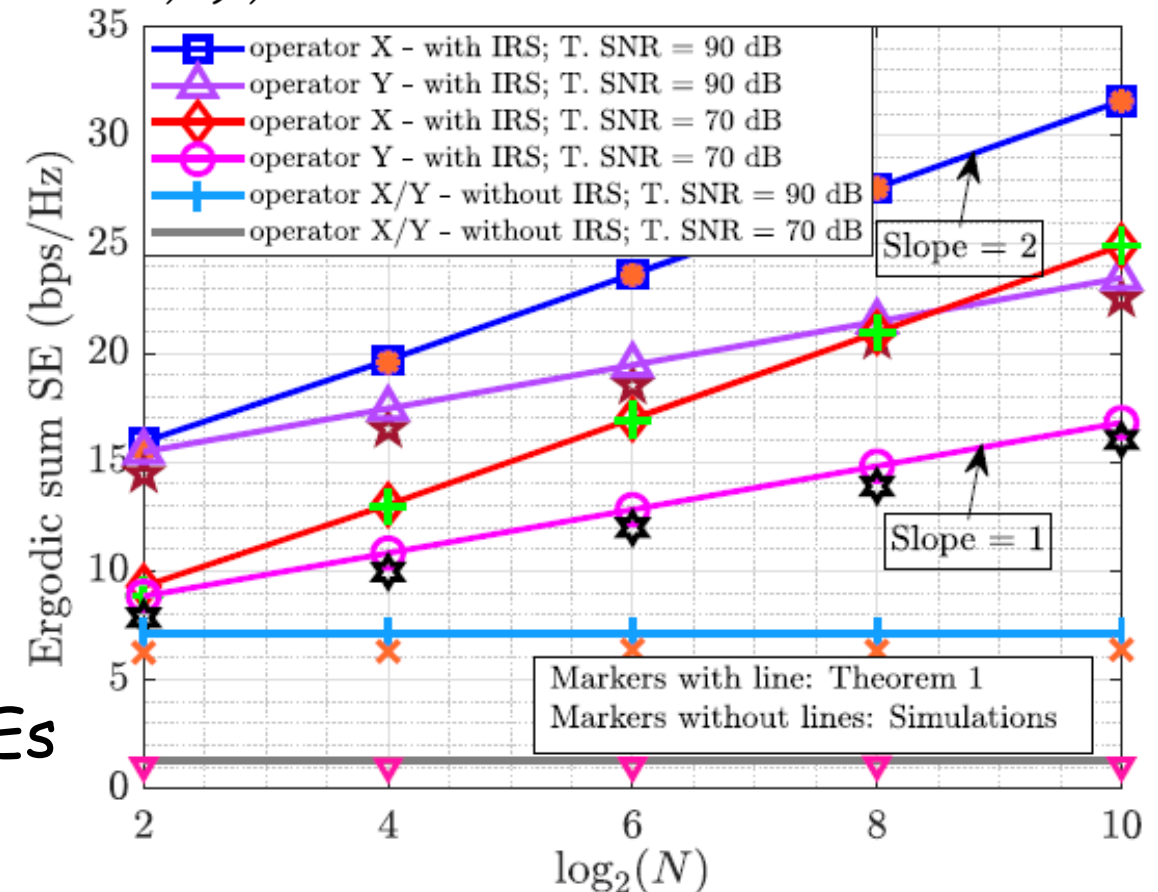
$$\bar{R}^{(X)} = \frac{1}{K} \sum_{k=1}^K \left(\mathcal{O} \left\{ \log_2 \left(1 + [N^2 c_{k,1} + N c_{k,2} + c_{k,3}] \frac{P}{\sigma^2} \right) \right\} \right)$$

and

$$\bar{R}^{(Y)} = \frac{1}{Q} \sum_{q=1}^Q \left(\mathcal{O} \left\{ \log_2 \left(1 + [N c_{q,1} + c_{q,2}] \frac{P}{\sigma^2} \right) \right\} \right)$$

SNR: N^2 for in-band & N for OOB UEs

- Acts as a scatterer @ OOB UEs
- Reception of multiple copies @ OOB UEs



Main Results - 2 (Instantaneous performance)

❑ **Outage Probability:** $P_{q,\text{out}}^\rho = \Pr(|h_q|^2 < \rho)$,

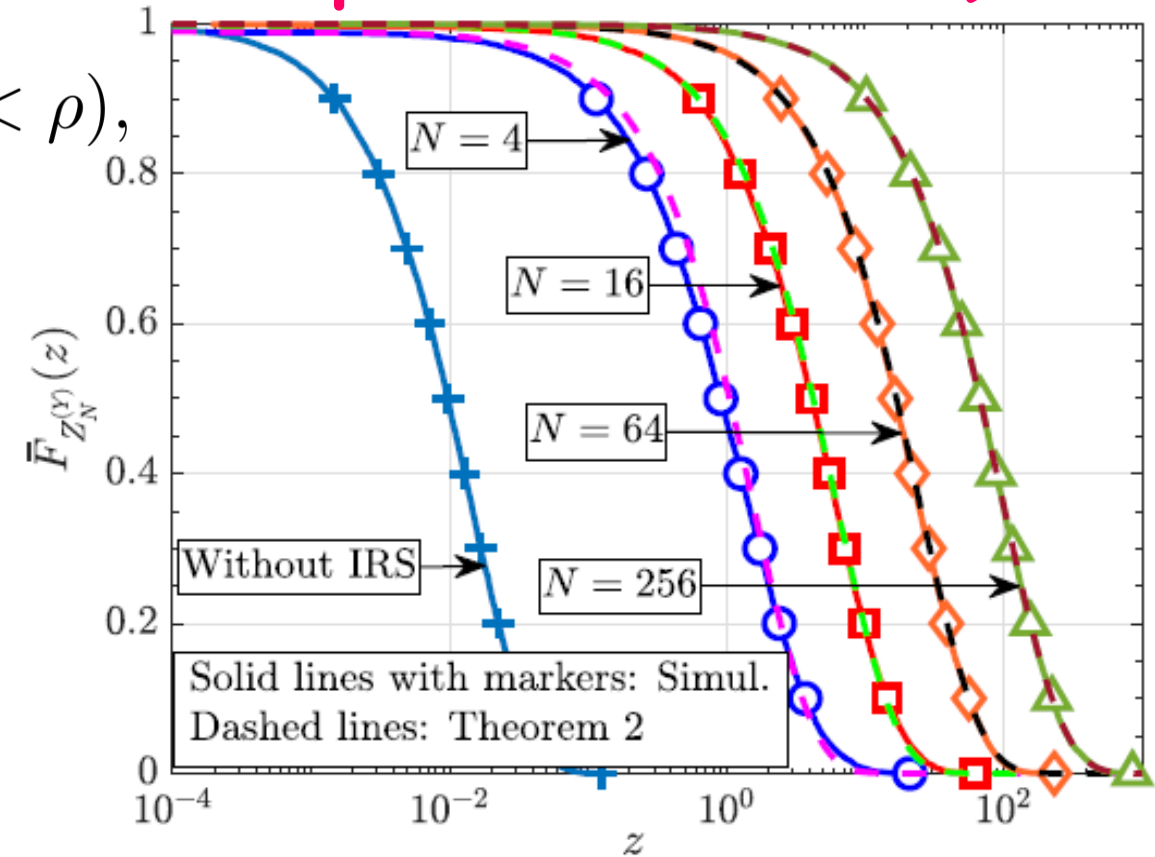
$$P_{q,\text{out}}^\rho = 1 - e^{-\frac{\rho}{N\beta_{r,q} + \beta_{d,q}}} \approx \frac{\rho}{N\beta_{r,q} + \beta_{d,q}}$$

❑ **Channel quality gain indicator:**

$$Z_N^{(Y)} \triangleq \left| \sum_{n=1}^N f_n^J g_{q,n} + h_{d,q} \right|^2 + (-\mathbf{1}_{\{N \neq 0\}}) |h_{d,q}|^2$$

➤ **CCDF of "CQ offset" @ OOB UE**

$$\bar{F}_{Z_N^{(Y)}}(z) = \begin{cases} 1 - \frac{1}{N\tilde{\beta} + 2} \times e^{\left(\frac{z}{\beta_{d,q}}\right)}, & \text{if } z < 0, \\ \left(\frac{N\tilde{\beta} + 1}{N\tilde{\beta} + 2}\right) \times e^{-\left(\frac{z}{\beta_{d,q}(1 + N\tilde{\beta})}\right)}, & \text{if } z \geq 0. \end{cases}$$



1. $\Pr\left(Z_N^{(Y)} < 0\right) = \mathcal{O}\left(\frac{1}{N}\right)$
2. $\{Z_N^{(Y)}\}_{N \cup \{0\}}$ - forms a **monotonic** sequence of **stochastically larger** random variables.

OOB Performance in mm-Wave Bands

- ❑ FR-2 Bands in the 5G standards
(24.25 GHz - 52.6 GHz, Rel. 15 , 2018)
- ❑ **Sparse** channels & **few** propagation **paths**
- ❑ Channels are highly **directional**!



System Modeling

- Directional **Saleh - Valenzuela** model:

$$\mathbf{f}^p = \sqrt{\frac{N}{L_{1,p}}} \sum_{i=1}^{L_{1,p}} \gamma_{i,p}^{(1)} \mathbf{a}_N^*(\phi_{i,p}); \quad \mathbf{g}_\ell = \sqrt{\frac{N}{L_{2,\ell}}} \sum_{j=1}^{L_{2,\ell}} \gamma_{j,\ell}^{(2)} z \mathbf{a}_N^*(\psi_{j,\ell}), \quad p \in \{X, Y\}$$

- Array steering** vector: $\mathbf{a}_N(\phi) = \frac{1}{\sqrt{N}} [1, e^{-j\pi\phi}, \dots, e^{-j(N-1)\pi\phi}]^T$

- Beam-resolution capability:

- N-element ULA** can form at most **N resolvable beams**

- Resolvable **beam book** and **angle book**:

$$\mathcal{A} \triangleq \{\mathbf{a}_N(\phi), \phi \in \Phi\}; \quad \Phi \triangleq \left\{ \left(-1 + \frac{2i}{N} \right) \middle| i = 0, \dots, N-1 \right\}$$

- Uniform** distribution over beam book:

$$\mathcal{U}_{\mathcal{A}}(\phi) = \frac{1}{|\Phi|} \mathbf{1}_{\{\phi \in \Phi\}} = \frac{1}{N} \mathbf{1}_{\{\phi \in \Phi\}}$$

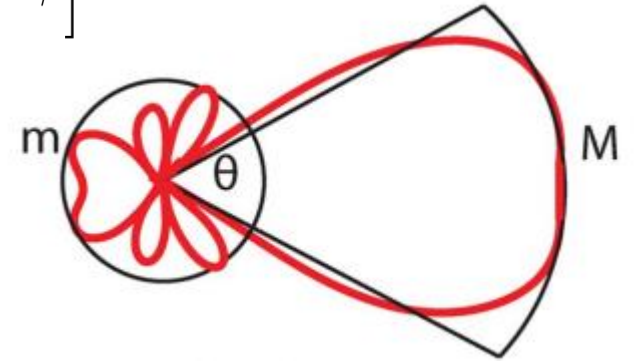


Fig. Directivity pattern of IRS

Categories of the study

LoS scenario

- ❖ IRS is **aligned** to the **dominant** cascaded path (virtual LoS)
- ❖ In-band UEs' **channel approximated** by the dominant path
- ❖ IRS vector is a **phasor**
- ❖ **Signaling** overhead does **NOT SCALE** with **N**

IRS optimization for **in-band** UEs in **mmWave** Bands

LoS scenarios

LoS + NLoS
scenarios
(L+)NLoS

(L+)NLoS scenario

- ❖ IRS is **jointly aligned** to all cascaded paths
- ❖ **No structure** in the IRS vector
- ❖ **Signaling** overhead **scales linearly** with IRS elements

WLOG, Channels for OOB UEs have general multipath modeling.

LoS scenarios

- Channel simplification for dominant path:
$$h_k = N\gamma_{1,X}^{(1)}\gamma_{1,k}^{(2)}\mathbf{a}_N^H(\psi_{1,k})\mathbf{\Theta}\mathbf{a}_N^*(\phi_{1,X}) + h_{d,k}$$
$$\stackrel{(a)}{=} N\left(\gamma_{1,X}^{(1)}\gamma_{1,k}^{(2)}\left(\mathbf{a}_N^H(\phi_{1,X}) \odot \mathbf{a}_N^H(\psi_{1,k})\right)\right)\boldsymbol{\theta} + h_{d,k},$$
$$= N\gamma_{X,k}\dot{\mathbf{a}}_N^H(\omega_{X,k}^1)\boldsymbol{\theta} + h_{d,k}$$
- Optimal IRS vector
$$\boldsymbol{\theta}^{\text{opt}} = \frac{h_{d,k}\gamma_{X,k}^*}{|h_{d,k}\gamma_{X,k}|} \times N\dot{\mathbf{a}}_N(\omega_{X,k}^1)$$
- IRS has a **uni-directional** response
@ in-band channel angle
- With probability $\frac{L}{N}$, OOB UE **benefits!**
- With probability $1 - \frac{L}{N}$, OOB perf. **same** as without an IRS

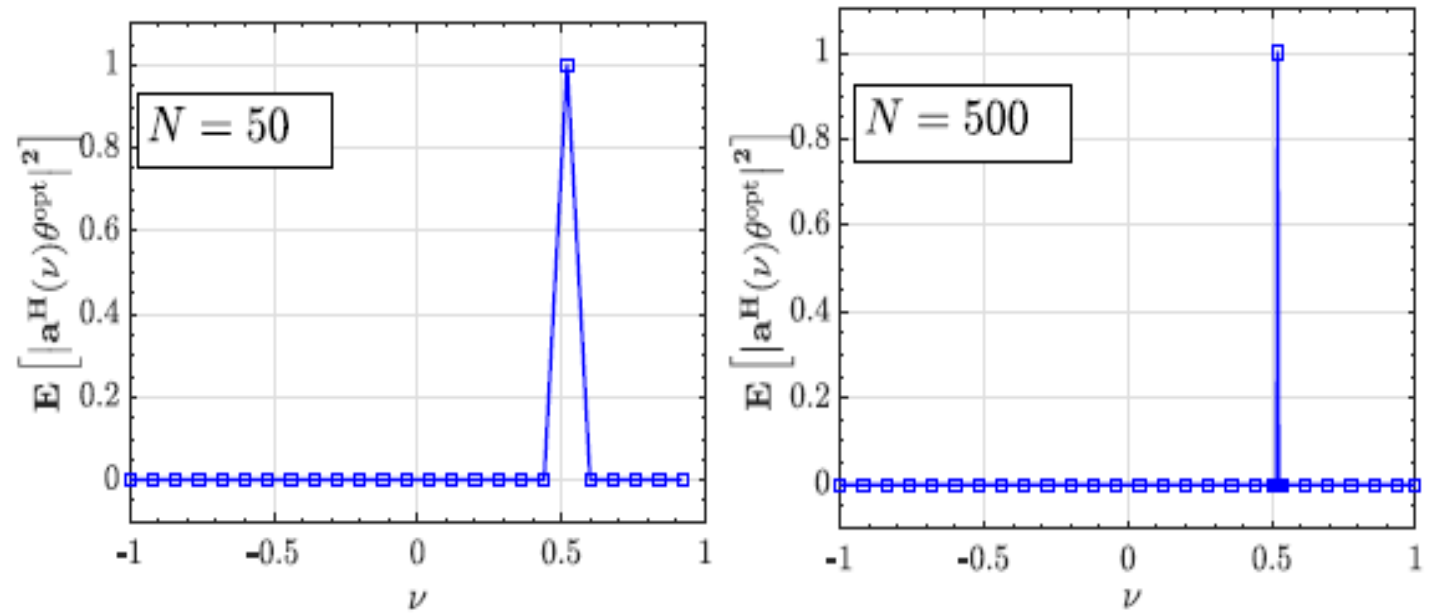


Fig. 3: Correlation response of the IRS vector and array steering vectors pointing at different spatial angles, ν , for (a) $N = 50$ and (b) $N = 500$. When $\nu = \omega_{X,k}^1$, the response attains its maximum value of 1.

Main Result - 1 (On Ergodic performance)

The ergodic sum SEs of operator **X** & **Y** scale as

$$\bar{S}_1^{(X)} \approx \frac{1}{K} \sum_{k=1}^K \log_2 \left(1 + \left[N^2 \beta_{r,k} + N \left(\frac{\pi^{3/2}}{4} \sqrt{\beta_{d,k} \beta_{r,k}} \right) + \beta_{d,k} \right] \frac{P}{\sigma^2} \right),$$

and

$$\bar{S}_1^{(Y)} \approx \frac{1}{Q} \sum_{q=1}^Q \left(\frac{\bar{L}}{N} \log_2 \left(1 + \left[\frac{N^2}{\bar{L}} \beta_{r,q} + \beta_{d,q} \right] \frac{P}{\sigma^2} \right) + \left(1 - \frac{\bar{L}}{N} \right) \log_2 \left(1 + \beta_{d,q} \frac{P}{\sigma^2} \right) \right),$$

where $\bar{L} \triangleq \min \{L, N\}$, and $L \triangleq L_1 L_2$

- ✓ IRS can **never degrade** the OOB performance
- ✓ **Linear** scaling of SNR with **N** is guaranteed if $L \geq N$.
- ✓ **Sub-linear** scaling of SNR with **N** if $L < N$.

The achievable OOB-SE in the **presence** of IRS is **at least** the SE in the **absence** of IRS.

Numerical results - 1 for LoS scenarios

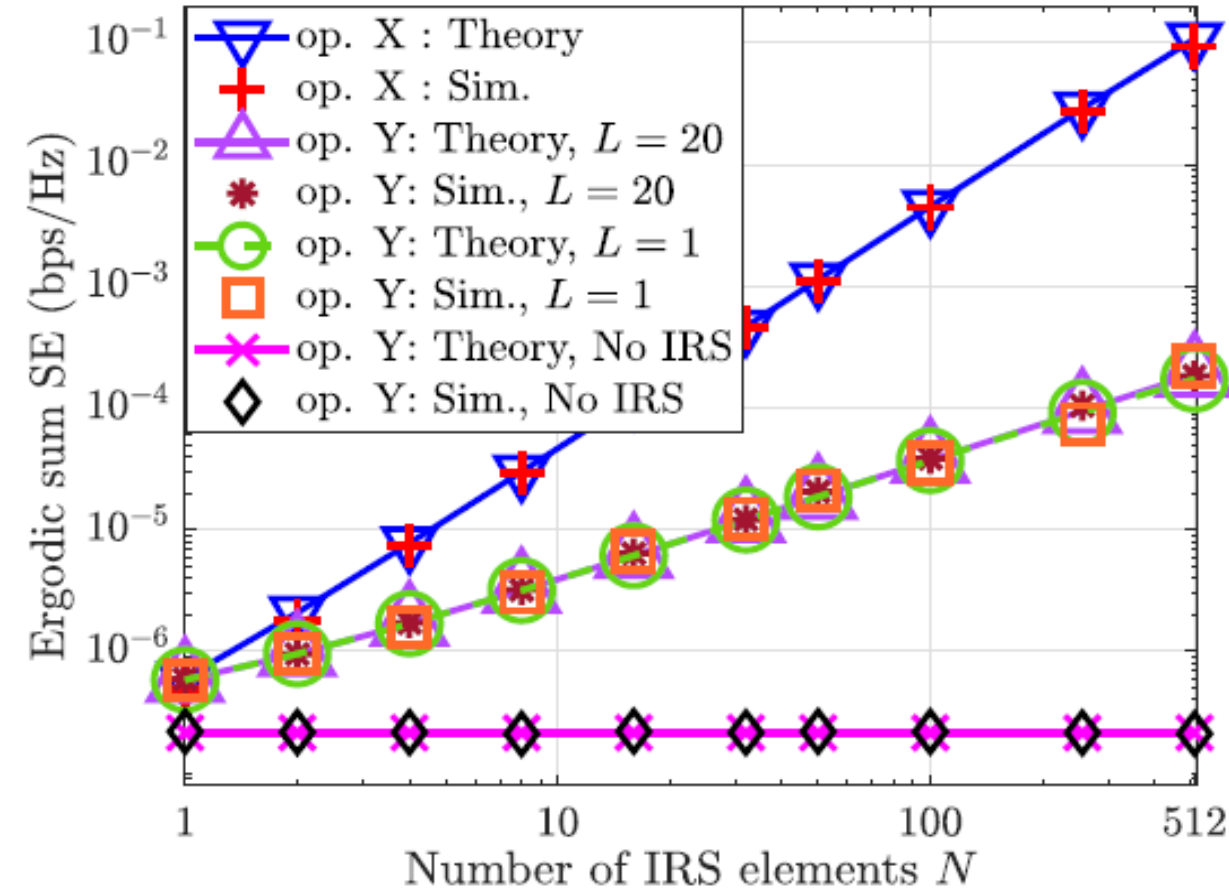


Fig. 6: Ergodic sum-SE vs. N in LoS scenarios at $\gamma = 70$ dB.

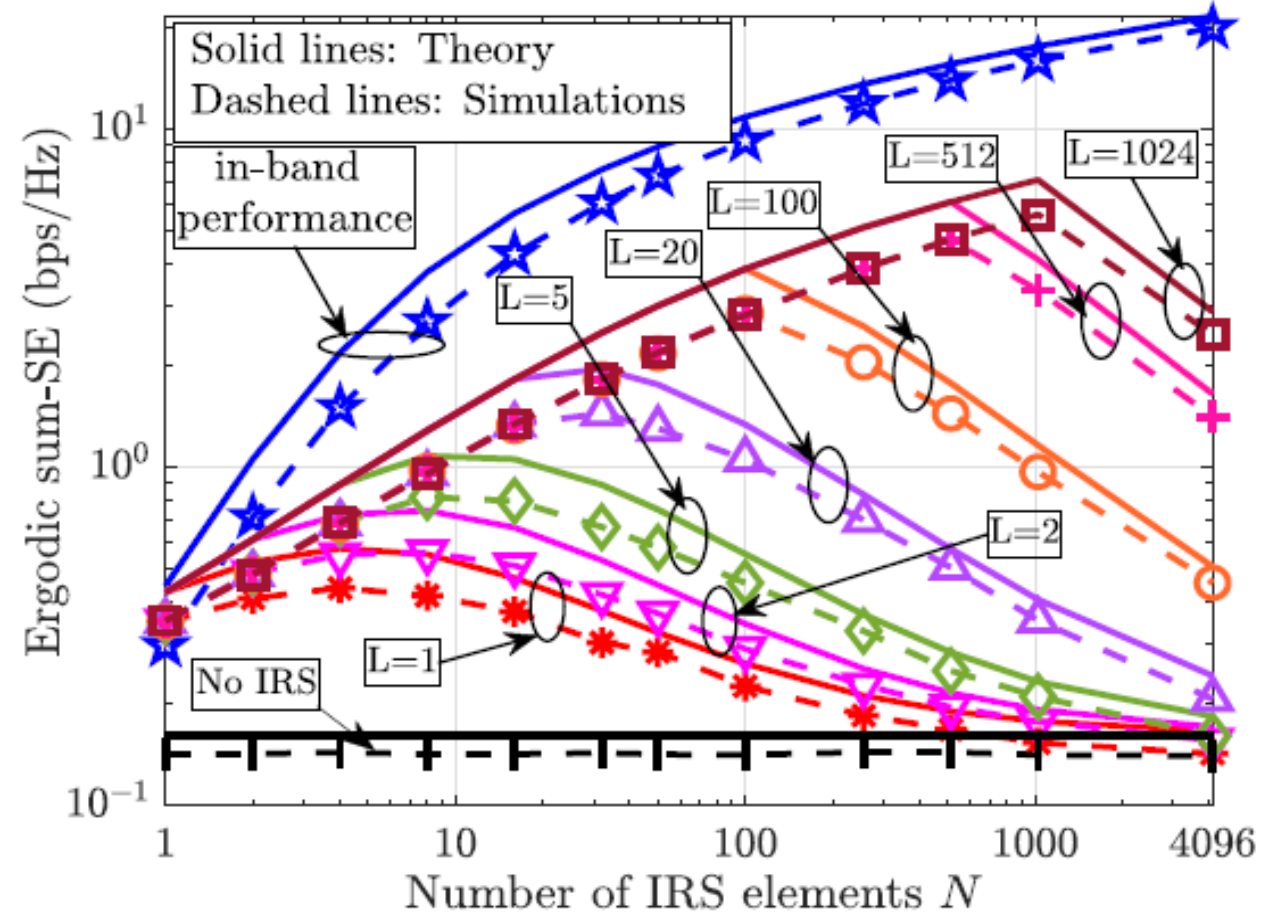


Fig. 7: Ergodic sum-SE vs. N in LoS scenarios at $\gamma = 130$ dB.

Main Results - 2 (On Instantaneous performance)

❑ Outage Probability:

$$F_{|h_q|^2}(\rho) \triangleq \Pr(|h_q|^2 < \rho) = 1 - e^{-\rho/\beta_{d,q}} - \frac{\bar{L}}{N} \left(\frac{\bar{L} e^{\frac{\bar{L}\beta_{d,q}}{N^2\beta_{r,q}}}}{N^2\beta_{r,q}} \mathcal{I}_0 \left(\rho; \beta_{d,q}, \frac{N^2}{\bar{L}}\beta_{r,q} \right) - e^{-\rho/\beta_{d,q}} \right),$$

where

$$\bar{L} = \min\{L, N\}, \text{ and } \mathcal{I}_0(x; c_1, c_2) = \int_{c_1}^{\infty} e^{\left\{ -\left(\frac{x}{t} + \frac{t}{c_2} \right) \right\}} dt.$$

❑ Channel quality gain indicator:

Define $G_1 = |h_q|^2$, and $G_0 = |h_q|^2$ in the *presence* and *absence* of the IRS, respectively. Then

$$\Pr(G_1 > \rho) \geq \Pr(G_0 > \rho) \quad \forall \rho \in \mathcal{R}^+,$$

The channel gain of the OOB-UE in the **presence** of an IRS **stochastically dominates** the channel gain in its **absence**.

Numerical results - 2 for LoS scenarios

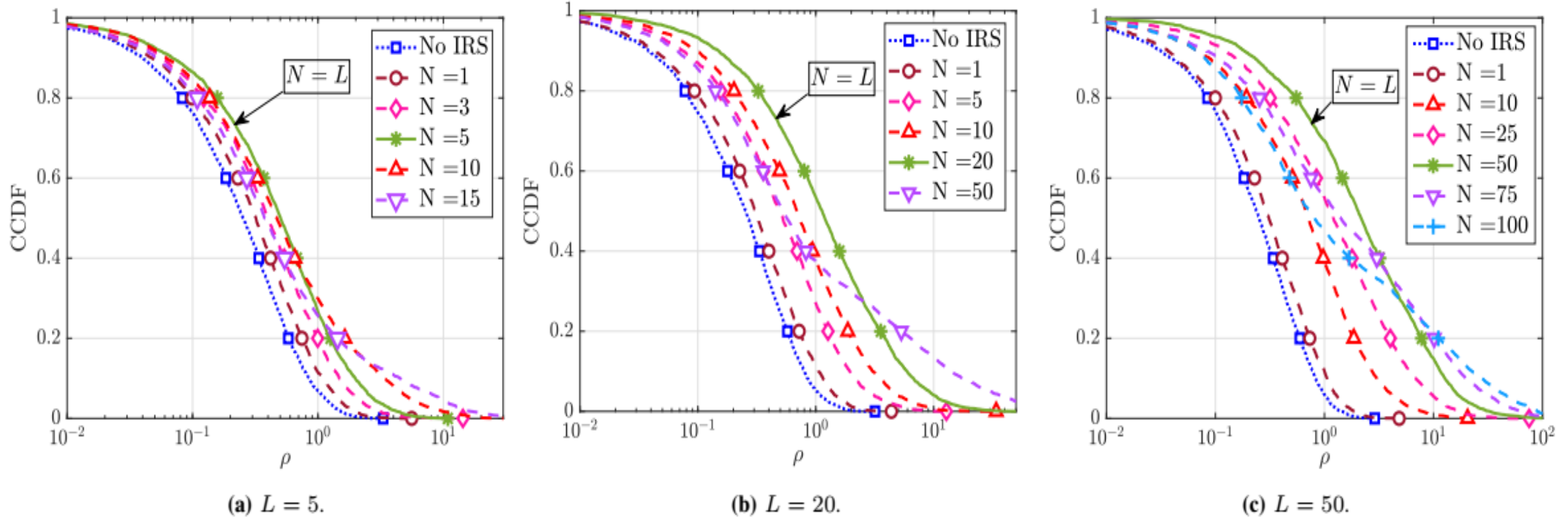


Fig. 8: CCDF of OOB UE's channel gain $|h_q|^2$ as a function of N in LoS scenarios of the mmWave bands at $\gamma = 130$ dB.

(L+)NLoS scenarios

- IRS is optimized to **LoS** and **NLoS** paths of the in-band UE's channel

- Channel: $h_k = h_{d,k} + \frac{N}{\sqrt{L}} \sum_{l=1}^L \gamma_{l,X}^{(1)} \gamma_{l,k}^{(2)} \dot{\mathbf{a}}_N^H(\omega_{X,k}^l) \boldsymbol{\theta}$.

- Optimal IRS configuration:

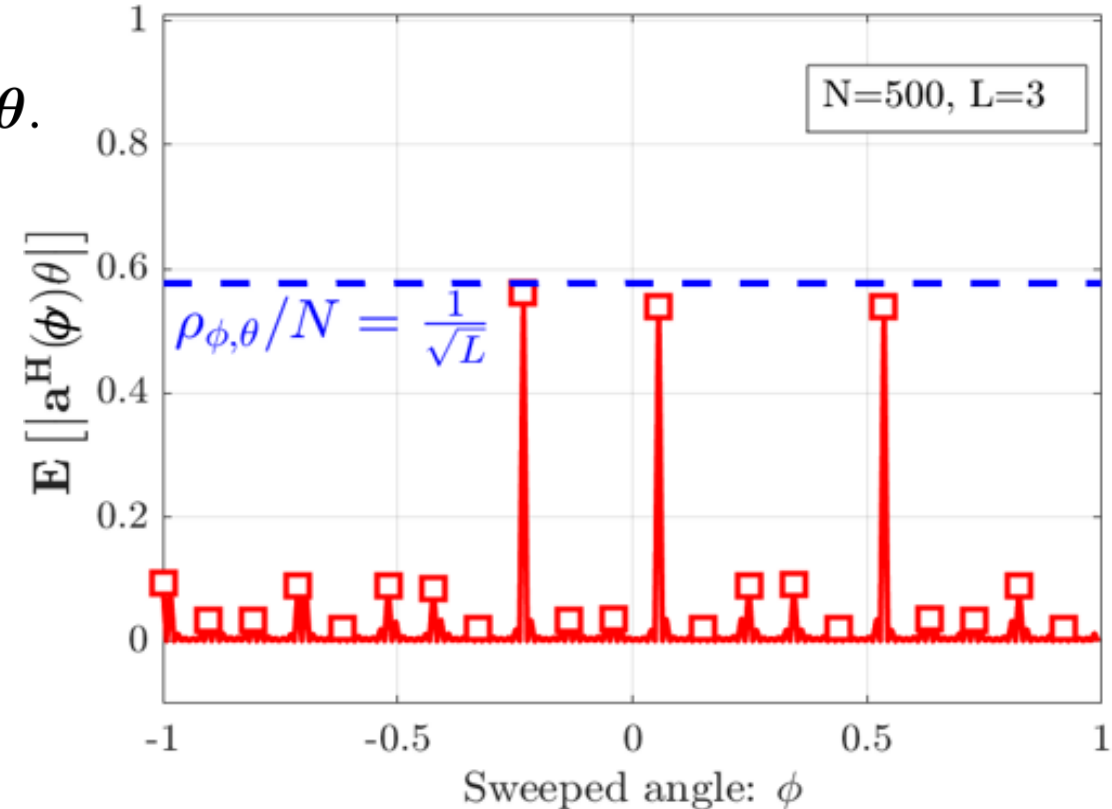
$$\boldsymbol{\theta}^{\text{opt}} = \frac{h_{d,k}}{|h_{d,k}|} \left(\sum_{l=1}^L \gamma_{l,X}^{(1)*} \gamma_{l,k}^{(2)*} \dot{\mathbf{a}}_N(\omega_{X,k}^l) \right) \odot \frac{1}{\left| \sum_{l=1}^L \gamma_{l,X}^{(1)} \gamma_{l,k}^{(2)} \dot{\mathbf{a}}_N(\omega_{X,k}^l) \right|},$$

- Directivity response:

Lemma 3. *The optimal IRS configuration*

has the following spatial amplitude response:

$$\rho_{\phi,\theta} = \begin{cases} \Omega \left(\frac{N}{\sqrt{L}} \right) + o(N), & \text{if } \phi \in \left\{ \omega_{X,k}^1, \dots, \omega_{X,k}^L \right\}, \\ o(N), & \text{if } \phi \in \Phi \setminus \left\{ \omega_{X,k}^1, \dots, \omega_{X,k}^L \right\}. \end{cases}$$



IRS distributes **energy**
without leakage

Main Result - 3 (On Ergodic performance)

The ergodic sum SEs of operator **X** & **Y** scale as

$$\bar{S}_L^{(X)} \approx \frac{1}{K} \sum_{k=1}^K \log_2 \left(1 + \left(N^2 \beta_{r,k} + N \sqrt{\pi \beta_{d,k} \beta_{r,k}} + \beta_{d,k} \right) \frac{P}{\sigma^2} \right),$$

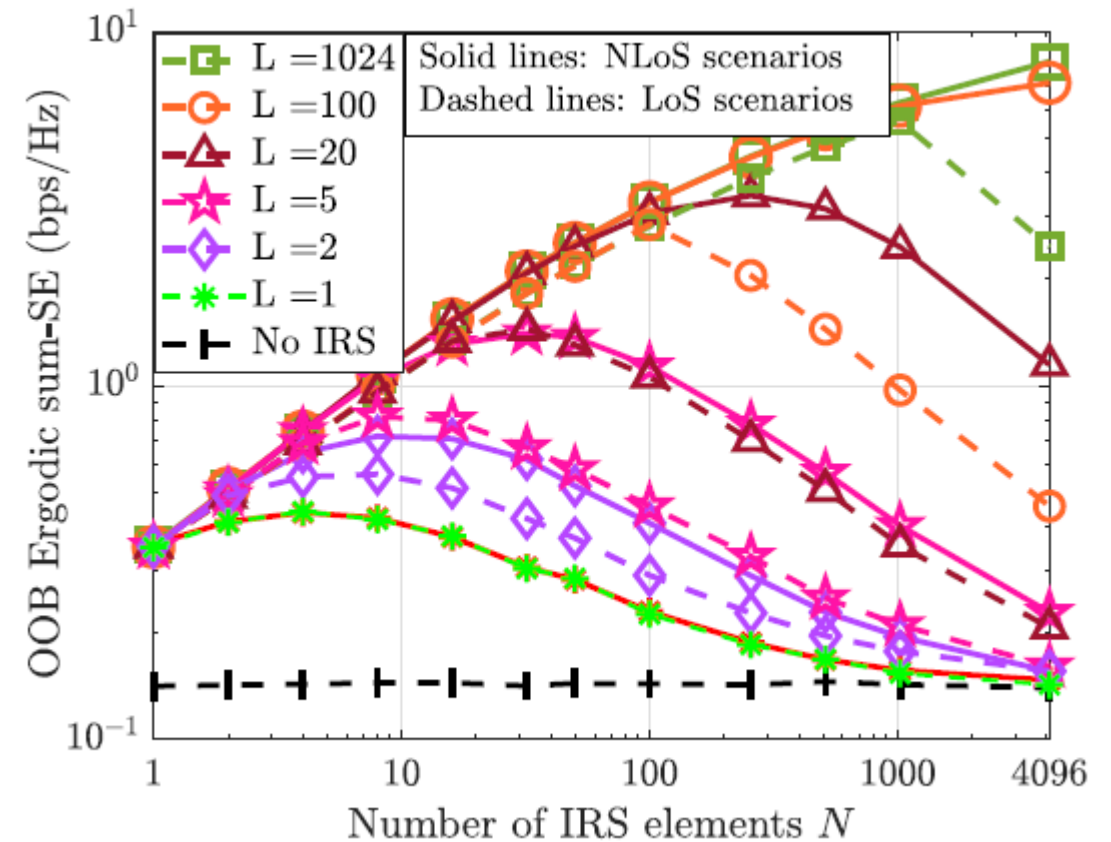
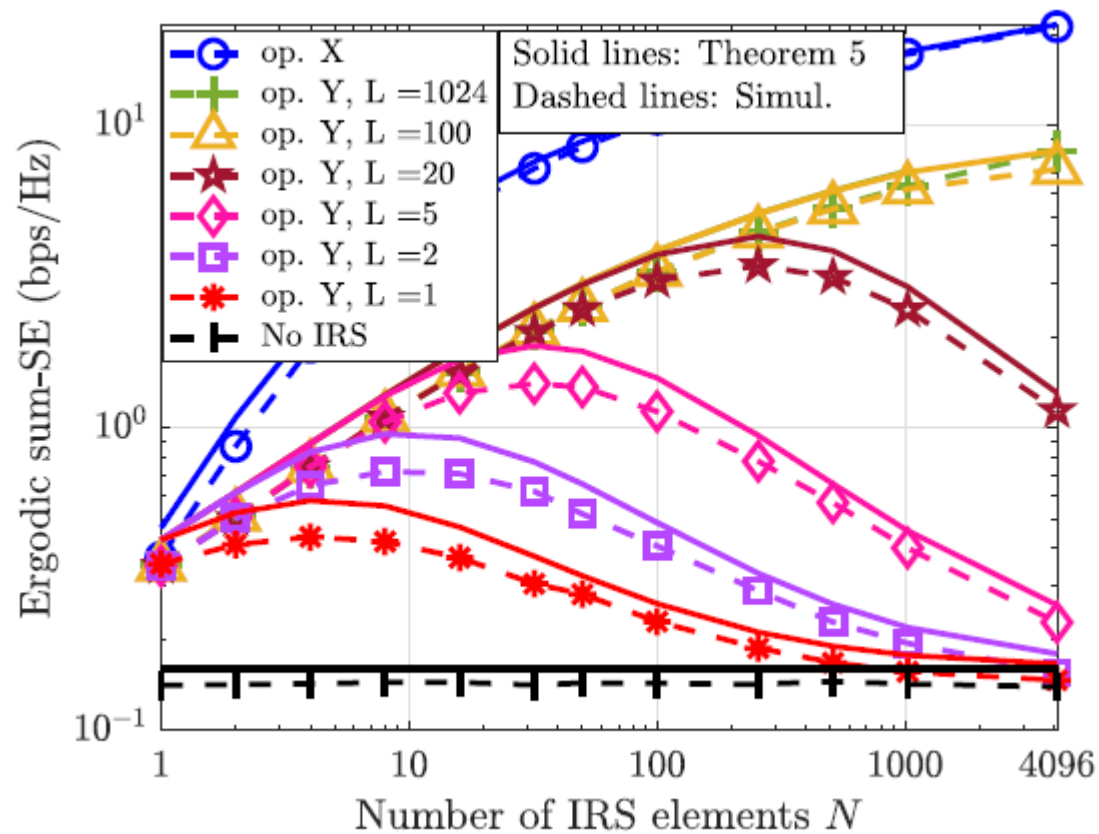
and

$$\bar{S}_L^{(Y)} \approx \begin{cases} \frac{1}{Q} \sum_{q=1}^Q \sum_{i=i_0}^L \frac{\binom{L}{i} \binom{N-L}{L-i}}{\binom{N}{L}} \times \log_2 \left(1 + \left(\beta_{d,q} + i \frac{N^2}{L^2} \beta_{r,q} \right) \frac{P}{\sigma^2} \right), & \text{if } L < N, \\ \frac{1}{Q} \sum_{q=1}^Q \log_2 \left(1 + (\beta_{d,q} + N \beta_{r,q}) \frac{P}{\sigma^2} \right), & \text{if } L \geq N. \end{cases}$$

where $i_0 \triangleq \max\{0, 2L - N\}$.

- ✓ IRS can **never degrade** the OOB performance
- ✓ **Linear** scaling of SNR with **N** is guaranteed if $L \geq \sqrt{N}$.
- ✓ **Sub-linear** scaling of SNR with **N** if $L < \sqrt{N}$.

Numerical results - 1 for (L+)NLoS scenarios



- Probability of 1-path alignment $\approx \frac{L^2}{N} > \frac{L}{N}$
- OOB perf. in NLOS is better than LOS
- Trade-off against feedback by in-band op.

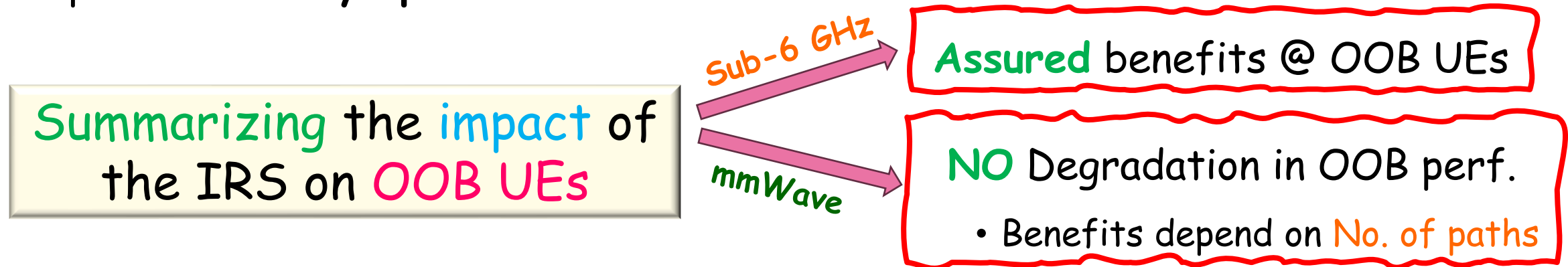
Key Inferences:

Sufficient L for linear scaling in LoS: N

Sufficient L for linear scaling in (L+)NLoS: $\sqrt{N}$

Concluding remarks

- IRS **enhances** the average **SNR** by N^2 at (in-band) UE of operator X
- Operator Y gets **enhancement** of average **SNR** by **N** for **free of cost**
 - Creates additional channel links to the OOB UEs: Reception of **multiple** copies
- **Scalable results**: **NO** operator will be **disadvantaged** due to IRS being controlled by one operator. **Every operator benefits!**



Thank You!

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